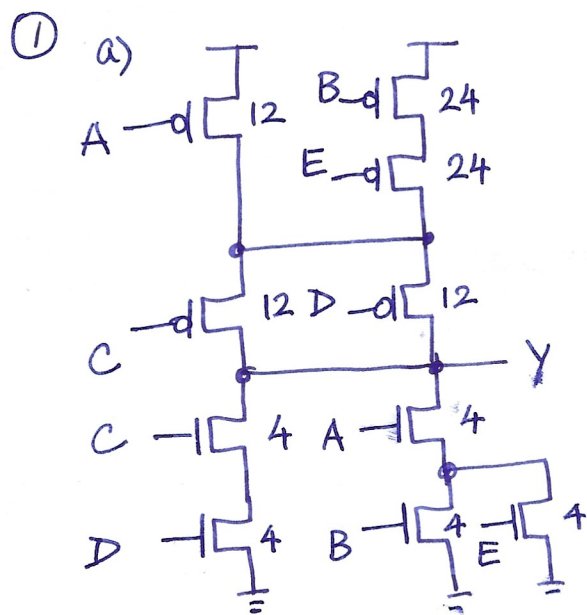




(b) $(10000) \rightarrow (10001)$
 $18.25 \tau_{inv}$

$(11011) \rightarrow (01011)$
 $7.75 \tau_{inv}$

② a) Propagation Delay = 10.52 ps

b) FO4 Delay = 26.3 ps

c) Duty cycle = 49.767%

③ Size of NAND2 = 4.

④ a) $\tau_{PHL} = 0.8845 \mu\text{s}$

b) $NM_H = NM_L = 1.0875 \text{ V}$

c) $\left(\frac{W}{L}\right)_n = 2$

a)

⑤ a) $T_{Cmin} = 250 \text{ ps}$; $F_{clkmax} = 4 \text{ GHz}$ ind. of width.

b) $t_{hold} \leq 100 \text{ ps}$; ind. of width

c) $t_{hold} \leq 160 \text{ ps}$; ind. of width

d) $T_{clkmin} = 410 \text{ ps}$; $F_{Cmax} = 2.44 \text{ GHz}$

$t_{hold} \leq 130 \text{ ps}$

⑥ a) Delay is Quadratic w/ l .

b) $l_{\max} = 73.30 \mu\text{m}$

⑦ Add 2 INV at the end $D_{\min} = 487.856 \text{ ps}$.

INV6 201.26 C

INV5 50.631 C

NAND4 28.022 C

NOR2 11.28 C

NAND2 7.945 C

⑧ $V_D = 0.125 \text{ V}$

$V_B = 0.7 \text{ V}$

$V_A = 1.275 \text{ V}$

$(W/L)_{1,2,3,4,5} = 64$

$(W/L)_6 = 16$